Configure Windows diagnostic data in your organization

Summarize this article for me

Applies to

- Windows 11 Enterprise
- Windows 11 Education
- Windows 11 Professional
- Windows 10 Enterprise
- Windows 10 Education
- Windows 10 Professional
- Windows Server 2016 and later
- Surface Hub
- HoloLens

This article describes the types of Windows diagnostic data sent back to Microsoft and the ways you can manage it within your organization. Microsoft uses the data to quickly identify and address issues affecting its customers.

Overview

Microsoft collects Windows diagnostic data to solve problems and to keep Windows up to date, secure, and operating properly. It also helps us improve Windows and related Microsoft products and services and, for customers who have turned on the **Tailored experiences** setting, to provide more relevant tips and recommendations to enhance Microsoft and third-party products and services for each customer's needs.

For more information about how Windows diagnostic data is used, see Diagnostics, feedback, and privacy in Windows.

Diagnostic data gives users a voice

Diagnostic data gives every user a voice in the operating system's development and ongoing improvement. It helps us understand how Windows behave in the real world, focus on user priorities, and make informed decisions that benefit both consumer and enterprise customers. The following sections offer real examples of these benefits.

Improve app and driver quality

Our ability to collect diagnostic data that drives improvements to Windows and Windows Server helps raise the bar for app and device driver quality. Diagnostic data helps us to quickly identify and fix critical reliability and security issues with apps and device drivers used on Windows. For example, we can identify an app that hangs on devices using a specific version of a video driver, allowing us to work with the app and device driver vendor to quickly fix the issue. The result is less downtime and reduced costs and increased productivity associated with troubleshooting these issues.

For example, in an earlier version of Windows there was a version of a video driver that was crashing on some devices, causing the device to restart. We detected the problem in our diagnostic data, and immediately contacted the third-party developer who builds the video driver. Working with the developer, we provided an updated driver to Windows Insiders within 24 hours. Based on diagnostic data from the Windows Insiders' devices, we were able to validate the new version of the video driver and rolled it out to the broad public as an update the next day. Diagnostic data helped us find, fix, and resolve this problem in just 48 hours, providing a better user experience and reducing costly support calls.

Improve end-user productivity

Windows diagnostic data also helps Microsoft better understand how customers use (or don't use) the operating system's features and related services. The insights we gain from this data helps us prioritize our engineering effort to directly impact our customers' experiences. These examples show how the use of diagnostic data enables Microsoft to build or enhance features which can help organizations increase employee productivity while lowering help desk calls.

- **Start menu.** How do people change the Start menu layout? Do they pin other apps to it? Are there any apps that they frequently unpin? We use this dataset to adjust the default Start menu layout to better reflect people's expectations when they turn on their device for the first time.
- **Cortana.** We use diagnostic data to monitor the scalability of our cloud service, improving search performance.
- **Application switching.** Research and observations from earlier Windows versions showed that people rarely used Alt+Tab to switch between apps. After discussing this with some users, we learned they loved the feature, saying that it would be highly productive, but they didn't know about it previously. Based on this, we created the Task View button in Windows to make this feature more discoverable. Later diagnostic data showed significantly higher usage of this feature.

How Microsoft handles diagnostic data

Use the following sections to learn more about how Microsoft handles diagnostic data.

Data collection

Depending on the diagnostic data settings on the device, diagnostic data can be collected via the following methods:

- Small payloads of structured information referred to as diagnostic data events, managed by the Connected User Experiences and Telemetry component.
- Diagnostic logs for additional troubleshooting, also managed by the Connected User Experiences and Telemetry component.
- Crash reporting and crash dumps, managed by Windows Error Reporting.

Later in this document we provide further details about how to control what's collected and what data can be included in these different types of diagnostic data.

As of March 6, 2024, Microsoft Edge diagnostic data is collected separately from Windows diagnostic data on Windows 10 (version 22H2 and newer) and Windows 11 (version 23H2 and newer) devices in the European Economic Area. The collection of Microsoft Edge diagnostic data is subject to its own settings. For more information related to this change, see Microsoft Edge, browsing data, and privacy.

Data transmission

All diagnostic data is encrypted using Transport Layer Security (TLS) and uses certificate pinning during transfer from the device to the Microsoft data management services.

Endpoints

The following table lists the endpoints related to how you can manage the collection and control of diagnostic data. For more information around the endpoints that are used to send data back to Microsoft, see the **Manage connection endpoints** section of the left-hand navigation menu.

Expand table

Windows service	Endpoint
Connected User Experiences and Telemetry	v10.events.data.microsoft.com v10c.events.data.microsoft.com v10.vortex-win.data.microsoft.com
Windows Error Reporting	watson.telemetry.microsoft.com umwatsonc.events.data.microsoft.com *-umwatsonc.events.data.microsoft.com

Windows service	Endpoint
	ceuswatcab01.blob.core.windows.net ceuswatcab02.blob.core.windows.net eaus2watcab01.blob.core.windows.net eaus2watcab02.blob.core.windows.net weus2watcab01.blob.core.windows.net weus2watcab02.blob.core.windows.net
Authentication	login.live.com IMPORTANT: This endpoint is used for device authentication. We don't recommend disabling this endpoint.
Online Crash Analysis	oca.telemetry.microsoft.com oca.microsoft.com kmwatsonc.events.data.microsoft.com *-kmwatsonc.events.data.microsoft.com
Settings	settings-win.data.microsoft.com IMPORTANT: This endpoint is required to remotely configure diagnostics-related settings and data collection. For example, we use the

Windows service	Endpoint
	settings endpoint to remotely block an event from being sent back to Microsoft, or to enroll a device in the Windows diagnostic data processor configuration. Don't block access to this endpoint. This endpoint doesn't upload Windows diagnostic data.

Proxy server authentication

If your organization uses proxy server authentication for internet access, make sure that it doesn't block the diagnostic data because of authentication.

Bypass (recommended)

Configure your proxy servers to not require proxy authentication for traffic to the diagnostic data endpoints. This option is the most comprehensive solution. It works for all versions of Windows 10 and Windows 11.

User proxy authentication

Configure devices to use the signed-in user's context for proxy authentication. This method requires the following configurations:

- Devices have the current quality update for a supported version of Windows
- Configure user-level proxy (WinINET proxy) in **Proxy settings** in the Network & Internet group of Windows Settings. You can also use the legacy Internet Options control panel.
- Make sure that the users have proxy permission to reach the diagnostic data endpoints. This option requires that the devices have console users with proxy permissions, so you can't use this method with headless devices.

> ⓘ **Important**

> The user proxy authentication approach is incompatible with the use of Microsoft Defender for Endpoint. This behavior is because this authentication relies on the **DisableEnterpriseAuthProxy** registry key set to `0`, while Microsoft Defender for Endpoint requires it to be set to `1`. For more information, see [Configure your devices to connect to the Defender for Endpoint service using a proxy](#).

Device proxy authentication

This approach supports the following scenarios:

- Headless devices, where no user signs in, or users of the device don't have internet access
- Authenticated proxies that don't use Windows Integrated Authentication
- If you also use Microsoft Defender for Endpoint

This approach is the most complex because it requires the following configurations:

- Make sure devices can reach the proxy server through WinHTTP in local system context. Use one of the following options to configure this behavior:
  - The command line `netsh winhttp set proxy`
  - Web proxy autodiscovery (WPAD) protocol
  - Transparent proxy
  - Configure device-wide WinINET proxy using the following group policy setting: **Make proxy settings per-machine (rather than per-user)** (ProxySettingsPerUser = `1`)
  - Routed connection, or that uses network address translation (NAT)

- Configure proxy servers to allow the computer accounts in Active Directory to access the diagnostic data endpoints. This configuration requires proxy servers to support Windows Integrated Authentication.

Data access

The principle of least privileged access guides access to Windows diagnostic data. Microsoft doesn't share personal data of our customers with third parties, except at the customer's discretion or for the limited purposes described in the Microsoft Privacy Statement. Microsoft may share business reports with hardware manufacturers and third-party partners that include aggregated and deidentified diagnostic data information. Data-sharing decisions are made by an internal team including privacy, legal, and data management.

Retention

Microsoft believes in and practices data minimization. We strive to gather only the info we need and to store it only for as long as it's needed to provide a service or for analysis. For more information, see the Microsoft Privacy Statement section on retention of data.

Diagnostic data settings

There are four diagnostic data collection settings. Each setting is described in more detail in the sections that follow.

- Diagnostic data off (Security)
- Required diagnostic data (Basic)
- Enhanced (This setting is only available on devices running Windows 10 version 1809 and earlier, Windows Server 2016, and Windows Server 2019.)
- Optional diagnostic data (Full)

Here's a summary of the types of data that is included with each setting:

Expand table

	Diagnostic data off (Security)	Required (Basic)	Enhanced	Optional (Full)
Diagnostic data events	No Windows diagnostic data sent.	Minimum data required to keep the device secure, up to date, and performing as expected.	Additional data about the websites you browse, how Windows and apps are used and how they perform, and device activity. The additional data helps Microsoft to fix and improve products and services for all users.	Additional data about the websites you browse, how Windows and apps are used and how they perform. This data also includes data about device activity, and enhanced error reporting that helps Microsoft to fix and improve products and services for all users.
Crash Metadata	N/A	Yes	Yes	Yes
Crash Dumps	N/A	No	Triage dumps only For more information about crash dumps, see Windows Error Reporting.	Full and triage memory dumps For more information about crash dumps, see Windows Error Reporting.
Diagnostic logs	N/A	No	No	Yes
Data collection	N/A	100%	Sampling applies	Sampling applies

Diagnostic data off

This setting was previously labeled as **Security**. When you configure this setting, no Windows diagnostic data is sent from your device. This is only available on Windows Server, Windows Enterprise, and Windows Education editions. If you choose this setting, devices in your organization will still be secure.

This was the default setting for Windows Server 2022 Datacenter: Azure Edition prior to December 13, 2022.

> **Note**
>
> If your organization relies on Windows Update, the minimum recommended setting is **Required diagnostic data**. Because no Windows Update information is collected when diagnostic data is off, important information about update failures isn't sent. Microsoft uses this information to fix the causes of those failures and improve the quality of our updates.

Required diagnostic data

Required diagnostic data, previously labeled as **Basic**, gathers a limited set of data that's critical for understanding the device and its configuration. This data helps to identify problems that can occur on a specific hardware or software configuration. For example, it can help determine if crashes are more frequent on devices with a specific amount of memory or that are running a specific driver version.

This is the default setting for Windows 10, version 1903 and later. Beginning December 13, 2022, it's also the default setting for Windows Server 2022 Datacenter: Azure Edition.

Required diagnostic data includes:

- Basic device data that helps provide an understanding about the types of Windows devices and the configurations and types of native and virtualized Windows Servers in the ecosystem. Examples include:
  - Device attributes, such as camera resolution and display type
  - Battery attributes, such as capacity and type
  - Networking attributes, such as number of network adapters, speed of network adapters, mobile operator network, and IMEI number
  - Processor and memory attributes, such as number of cores, architecture, speed, memory size, and firmware
  - Virtualization attribute, such as Second Level Address Translation (SLAT) support and guest operating system
  - Operating system attributes, such as Windows edition and virtualization state
  - Storage attributes, such as number of drives, type, and size

- Quality metrics that help provide an understanding about how the Connected User Experiences and diagnostic data component is functioning, including % of uploaded events, dropped events, blocked events, and the last upload time.
- Quality-related information that helps Microsoft develop a basic understanding of how a device and its operating system are performing. Some examples are the device characteristics of a Connected Standby device, the number of crashes or hangs, and app state change details, such as how much processor time and memory were used, and the total uptime for an app.
- Compatibility data that helps provide an understanding about which apps are installed on a device or virtual machine and identifies potential compatibility problems.
- System data that helps provide an understanding about whether a device meets the minimum requirements to upgrade to the next version of the operating system. System information includes the amount of memory, as well as information about the processor and BIOS.
- A list of accessory device data, such as printers or external storage devices, that are connected to Windows devices and whether these devices will function after upgrading to a new version of the operating system.
- Driver data that includes specific driver activity that's meant to help figure out whether apps and devices will function after upgrading to a new version of the operating system. This can help to determine blocking issues and then help Microsoft and our partners apply fixes and improvements.
- Information about how the Microsoft Store performs, including app downloads, installations, and updates. It also includes Microsoft Store launches, page views, suspend and resumes, and obtaining licenses.

Enhanced diagnostic data

In Windows 10 (version 1809 and earlier) and Windows Server 2019, enhanced diagnostic data includes data about the websites you browse, how Windows and apps are used and how they perform, and device activity. The additional data helps Microsoft to fix and improve products and services for all users.

> **Important**
>
> This diagnostic data setting is not available on Windows 11 and Windows Server 2022 and has been replaced with policies that can control the amount of optional diagnostic data that is sent. More information on these settings are available in the Manage diagnostic data using Group Policy and MDM section of this article.

When you choose to send enhanced diagnostic data, required diagnostic data will always be included, and we collect the following additional information:

- Operating system events that help to gain insights into different areas of the operating system, including networking, Hyper-V, Cortana, storage, file system, and other components.
- Operating system app events resulting from Microsoft apps and management tools that were downloaded from the Microsoft Store or pre-installed with Windows or Windows Server, including Server Manager, Photos, Mail, and Microsoft Edge.
- Device-specific events that are specific to certain devices, such as Surface Hub and Microsoft HoloLens. For example, Microsoft HoloLens sends Holographic Processing Unit (HPU)-related events.
- All crash dump types, except for heap dumps and full dumps. For more information about crash dumps, see Windows Error Reporting.

Optional diagnostic data

Optional diagnostic data, previously labeled as **Full**, includes more detailed information about your device and its settings, capabilities, and device health. Optional diagnostic data also includes data about the websites you browse, device activity, and enhanced error reporting that helps Microsoft to fix and improve products and services for all users. When you choose to send optional diagnostic data, required diagnostic data will always be included, and we collect the following additional information:

- Additional data about the device, connectivity, and configuration, beyond that collected under required diagnostic data.

- Status and logging information about the health of operating system and other system components beyond what is collected under required diagnostic data.
- App activity, such as which programs are launched on a device, how long they run, and how quickly they respond to input.
- Browser activity, including browsing history and search terms, in Microsoft browsers (Microsoft Edge or Internet Explorer).
- Enhanced error reporting, including the memory state of the device when a system or app crash occurs (which may unintentionally contain user content, such as parts of a file you were using when the problem occurred). Crash data is never used for Tailored experiences.

> **Note**
>
> Crash dumps collected in optional diagnostic data may unintentionally contain personal data, such as portions of memory from a document and a web page. For more information about crash dumps, see Windows Error Reporting.

Manage diagnostic data using Group Policy and MDM

Use the steps in this section to configure the diagnostic data settings for Windows and Windows Server in your organization.

> **Important**
>
> - These diagnostic data settings only apply to components, features, and apps that are considered a part of the Windows operating system.
> - Third-party apps and other Microsoft apps, such as Microsoft 365 Apps, that customers install may also collect and send diagnostic data using their own controls.
> - You should work with your app vendors to understand their diagnostic data policy, and how you can opt in or opt out.

- For more information on how Microsoft 365 Apps uses diagnostic data, see Overview of privacy controls for Microsoft 365 Apps for enterprise.
- If you would like to control Windows data collection that isn't Windows diagnostic data, see Manage connections from Windows operating system components to Microsoft services.

You can configure your device's diagnostic data settings using the management tools you're already using, such as Group Policy or MDM.

Use the appropriate value in the table below when you configure the management policy.

Expand table

Category	Value
Diagnostic data off (Security)	0
Required (Basic)	1
Enhanced	2
Optional (Full)	3

Note

When both the Computer Configuration policy and User Configuration policies are set, the more restrictive policy is used.

Use Group Policy to manage diagnostic data collection

You can use Group Policy to set your organization's diagnostic data setting:

1. From the Group Policy Management Console, go to **Computer Configuration** > **Administrative Templates** > **Windows Components** > **Data Collection and Preview Builds**.

2. Double-click **Allow diagnostic data** (or **Allow telemetry** on Windows 10 version 1809 and earlier, and on Windows Server 2019).

> ⓘ **Note**
>
> If devices in your organization are running Windows 10, 1803 and later, the user can still use Settings to set the diagnostic data setting to a more restrictive value, unless the **Configure diagnostic data opt-in settings user interface** policy is set.

3. In the **Options** box, choose the setting that you want to configure, and then select **OK**.

Use Group Policy to manage optional diagnostic data collection

The following policy lets you limit the types of crash dumps that can be sent back to Microsoft. If this policy is enabled, Windows Error Reporting will send only kernel mini dumps and user mode triage dumps.

1. From the Group Policy Management Console, go to **Computer Configuration** > **Administrative Templates** > **Windows Components** > **Data Collection and Preview Builds**.

2. Double-click **Limit dump collection**.

3. In the **Options** box, choose the setting that you want to configure, and then select **OK**.

You can also limit the number of diagnostic logs that are sent back to Microsoft. If this policy is enabled, diagnostic logs aren't sent back to Microsoft.

1. From the Group Policy Management Console, go to **Computer Configuration** > **Administrative Templates** > **Windows Components** > **Data Collection and Preview Builds**.

2. Double-click **Limit diagnostic log collection**.

3. In the **Options** box, choose the setting that you want to configure, and then select **OK**.

Use MDM to manage diagnostic data collection

Use Policy Configuration Service Provider (CSP) to apply the following MDM policies:

- System/AllowTelemetry
- System/LimitDumpCollection
- System/LimitDiagnosticLogCollection

> **Note**
>
> The last two policies are only available on Windows 11 and Windows Server 2022.

Enable Windows diagnostic data processor configuration

The Windows diagnostic data processor configuration enables you to be the controller, as defined by the European Union General Data Protection Regulation (GDPR), for the Windows diagnostic data collected from your Windows devices that meet the configuration requirements.

Prerequisites

- Use a supported version of Windows 10 or Windows 11
- The following editions are supported:
  - Enterprise

  - Professional
  - Education
- The device must be joined to Microsoft Entra (can be a Microsoft Entra hybrid join).

> **Note**
>
> In all cases, enrollment in the Windows diagnostic data processor configuration requires a device to be joined to a Microsoft Entra tenant. If a device isn't properly enrolled, Microsoft will act as the controller for Windows diagnostic data in accordance with the Microsoft Privacy Statement and the Data Protection Addendum terms won't apply.

For the best experience, use the most current build of any operating system specified above. Configuration functionality and availability may vary on older systems. For release information, see Windows 10 Enterprise and Education and Windows 11 Enterprise and Education on the Microsoft Lifecycle Policy site.

The diagnostic data setting on the device should be set to Required diagnostic data or higher, and the following endpoints need to be reachable:

- us-v10c.events.data.microsoft.com (eu-v10c.events.data.microsoft.com for tenants with billing address in the EU Data Boundary)
- watsonc.events.data.microsoft.com (eu-watsonc.events.data.microsoft.com for tenants with billing address in the EU Data Boundary)
- settings-win.data.microsoft.com
- *.blob.core.windows.net

Tenants with billing addresses in countries or regions in the Middle East and Africa, as well as European countries or regions not in the EU, also use the eu-v10c.events.data.microsoft.com and eu-watsonc.events.data.microsoft.com endpoints. Their diagnostic data is processed initially in Europe, but those tenants aren't considered part of the EU Data Boundary.

> **Note**

- Windows diagnostic data collected from a device before it was enabled with Windows diagnostic data processor configuration will be deleted when this configuration is enabled.
- When you enable devices with the Windows diagnostic data processor configuration, users may continue to submit feedback through various channels such as Windows feedback hub or Edge feedback. However, the feedback data is not subject to the terms of the Windows diagnostic data processor configuration. If this is not desired, we recommend that you disable feedback using the available policies or application management solutions.

Enabling Windows diagnostic data processor configuration

> **Note**
>
> The information in this section applies to the following versions of Windows:
>
> - Windows 10, versions 20H2 and newer
> - Windows 11, versions 21H2 and newer

Starting with the January 2023 preview cumulative update, how you enable the processor configuration option depends on the billing address of the Microsoft Entra tenant to which your devices are joined.

Devices in Microsoft Entra tenants with a billing address in the European Union (EU) or European Free Trade Association (EFTA)

For Windows devices with diagnostic data turned on and that are joined to an Microsoft Entra tenant with billing address in the EU or EFTA, the Windows diagnostic data for that device will be automatically configured for the processor option. The Windows diagnostic data for those devices will be processed in Europe.

From a compliance standpoint, this change means that Microsoft will be the processor and the organization will be the controller of the Windows diagnostic data. IT admins for those organizations will become responsible for responding to their users' data subject requests.

Devices in Microsoft Entra tenants with a billing address outside of the EU and EFTA

For Windows devices with diagnostic data turned on and that are joined to an Microsoft Entra tenant with billing address outside of the EU and EFTA, to enable the processor configuration option, the organization must sign up for any of the following enterprise services, which rely on diagnostic data:

- Windows Update for Business reports
- Windows Autopatch
- Windows updates reports (in Microsoft Intune)

(Additional licensing requirements may apply to use these services.)

If you don't sign up for any of these enterprise services, Microsoft will act as controller for the diagnostic data.

Enabling Windows diagnostic data processor configuration on older versions of Windows

> **Note**
>
> The information in this section applies to the following versions of Windows:
>
> - Windows 10, versions 1809, 1903, 1909, and 2004.
> - Newer versions of Windows 10 and Windows 11 that haven't updated yet to at least the January 2023 preview cumulative update.

To enable Windows diagnostic data processor configuration, you can use Group Policy or a custom setting in an MDM solution, such as Microsoft Intune.

- For Group Policy, you can use the "Allow commercial data pipeline" policy, which is also available in the Intune settings catalog.
- For an MDM solution, you can use the AllowCommercialDataPipeline setting in the System Policy configuration service provider (CSP).

For more information about AllowCommercialDataPipeline and the "Allow commercial data pipeline" policy, review this information.

Change privacy settings on a single server

You can also change the privacy settings on a server running either the Azure Stack HCI operating system or Windows Server. For more information, see Change privacy settings on individual servers.

Last updated on 09/11/2025

网络连接状态指示器 （NCSI） 故障排除指南

适用于: Supported versions of Windows Server

本指南旨在帮助排查网络连接状态指示器（NCSI）问题。

NCSI 基础知识

网络位置感知（NLA）服务确定 Windows 具有哪种类型的网络连接。

在 NLA 服务在 Windows 10 中加载 *ncsi.dll文件之前* ，它将执行操作并接收有关网络的通知和状态信息。 NCSI 评估的连接状态由各种应用程序使用，例如Microsoft Outlook、Microsoft Teams、Skype、Windows 更新、Microsoft DirectAccess 和某些第三方软件。

从 Windows 11 开始，作业由网络列表服务（netprofm）服务执行。

NCSI 主要通过网络探测运行，这些探测涉及将简单的网络请求发送到终结点并等待响应。

活动探测

活动探测可以涉及对特定 NCSI 地址的域名系统（DNS）查找或定向到 Web 探测服务器的请求。 虽然服务器由 Internet 上的Microsoft托管，但它也可以是其企业中的客户专用探测服务器，尽管这种情况并不常见。

如果 NCSI 发送探测并接收有效的响应，计算机将连接到 Internet。

> **注意**
>
> 不要禁用活动探测来解决问题。

被动探测

虽然主动探测显式执行与网络相关的操作来获取有关网络状态的信息，但被动探测使用从接收的数据中学到的信息来实现相同的目的。

它依赖于网络统计信息，例如最近发送或接收的数据包、这些帧的生存时间（TTL）值，以及来自其他组件的通知。

故障排除清单

1. 网络状态任务栏图标

网络状态任务栏图标是网络连接的基本指示。 它会通知用户网络是完全可用的，还是存在某种程度的网络问题。

- 通过 Wi-Fi 进行完全 Internet 访问：
- 通过以太网进行完全 Internet 访问： 或
- 某种程度的网络中断：
  - **未连接 - 连接可用**：
  - **识别 无 Internet 访问**： 、 或

> **注意**
>
> 不完全依赖指标是有益的。 NCSI 无法完成探测并不一定意味着客户端计算机无法访问 Internet。

若要解决 NCSI 问题，请确保 NCSI 可以将活动探测发送到 Internet 并成功接收响应。

2. NCSI 浏览器测试

打开浏览器并访问 Microsoft连接测试 （`ipv6.msftconnecttext.com` 适用于 IPv6）。 预期内容为“Microsoft连接测试”。

如果无法获取文本文件，请验证以下项：

- 代理设置在目标客户端上正确配置。
- 代理服务器不会限制对上述地址的访问。

如果客户端在 Windows 10 版本 1607 之前，请使用 Microsoft NCSI （`ipv6.msftncsi.com` 适用于 IPv6）。 预期内容是包含内容“Microsoft NCSI”的纯文本文件。

有时，当 NCSI 失败时，浏览器上的手动测试会成功，因为它绕过代理。 在这种情况下，通过网络跟踪跟踪跟踪流量有助于识别异常行为。

3.检查 NCSI 事件日志

检查路径中的 NCSI 事件日志： *应用程序和服务日志\Microsoft\Windows\NCSI\Operational*。 例如：

4.检查注册表位置是否有任何更改

> ⓘ **注意**
>
> 必须启用活动探测。

HTTP Web 探测服务器、路径、预期探测内容以及 DNS 探测主机和内容在注册表路径下预定义：

`HKLM\SYSTEM\CurrentControlSet\Services\NlaSvc\Parameters\Internet`

Windows 10 版本 1607 及更高版本中的默认值：

展开表

名称	类型	数据
（默认值）	REG_SZ	(未设置值)
ActiveDnsProbeContent	REG_SZ	131.107.255.255
ActiveDnsProbeContentV6	REG_SZ	fd3e：4f5a：5b81：：1
ActiveDnsProbeHost	REG_SZ	dns.msftncsi.com
ActiveDnsProbeHostV6	REG_SZ	dns.msftncsi.com
ActiveWebProbeContent	REG_SZ	Microsoft连接测试
ActiveWebProbeContentV6	REG_SZ	Microsoft连接测试
ActiveWebProbeHost	REG_SZ	www.msftconnecttest.com
ActiveWebProbeHostV6	REG_SZ	ipv6.msftconnecttest.com
ActiveWebProbePath	REG_SZ	connecttest.txt
ActiveWebProbePathV6	REG_SZ	connecttest.txt
CaptivePortalTimer	REG_DWORD	0x00000000 （0）

名称	类型	数据
CaptivePortalTimerBackOffincrementsInSeconds	REG_DWORD	0x00000005 （5）
CaptivePortalTimerMaxInSeconds	REG_DWORD	0x0000001e （30）
EnableActiveProbing	REG_DWORD	0x00000001 （1）
PassivePollPeriod	REG_DWORD	0x0000000f （15）
StaleThreshold	REG_DWORD	0x0000001e （30）
WebTimeout	REG_DWORD	0x00000023 （35）

Windows 10 版本 1511、Windows 10 版本 1507、Windows 8.1 和 Windows 8 中的默认值：

展开表

名称	类型	数据
（默认值）	REG_SZ	(未设置值)
ActiveDnsProbeContent	REG_SZ	131.107.255.255
ActiveDnsProbeContentV6	REG_SZ	fd3e：4f5a：5b81：：1
ActiveDnsProbeHost	REG_SZ	dns.msftncsi.com
ActiveDnsProbeHostV6	REG_SZ	dns.msftncsi.com
ActiveWebProbeContent	REG_SZ	Microsoft NCSI
ActiveWebProbeContentV6	REG_SZ	Microsoft NCSI
ActiveWebProbeHost	REG_SZ	www.msftncsi.com
ActiveWebProbeHostV6	REG_SZ	ipv6.msftncsi.com
ActiveWebProbePath	REG_SZ	ncsi.txt
ActiveWebProbePathV6	REG_SZ	ncsi.txt
EnableActiveProbing	REG_DWORD	0x00000001 （1）
PassivePollPeriod	REG_DWORD	0x0000000f （15）
StaleThreshold	REG_DWORD	0x0000001e （30）
WebTimeout	REG_DWORD	0x00000023 （35）

Windows 7 和早期版本没有此功能。

从 Windows 10 版本 1607 开始，Web 探测 （HTTP） 请求将发送到 Microsoft连接测试。 预期响应为“HTTP 200 正常”，有效负载包含“Microsoft连接测试”。

> **注意**
>
> 如果通过本地注册表项禁用活动探测，则值为 `EnableActiveProbing` *0*。 请确保将值设置为 *1*。

5. 连接简单测试

运行以下 PowerShell cmdlet 以测试连接性：

PowerShell

```
Get-NetConnectionProfile
```

下面是良好连接的示例：

输出

```
Name                     : XYZ
InterfaceAlias           : Ethernet
InterfaceIndex           : 5
NetworkCategory          : Private
DomainAuthenticationKind : None
IPv4Connectivity         : Internet
IPv6Connectivity         : Internet
```

如果任何接口的 IPv4 或 IPv6 连接指示 `Internet`，则计算机的连接被视为 Internet。 在这种情况下，此问题当前不可重现或与 NCSI 无关。

6.检查哪些探测用于检测受影响计算机上的 Internet

根据各种因素，计算机可能使用特定类型的活动探测来确定 Internet 连接。 此外，考虑到主动和被动探测都处于活动状态，使用主动探测时计算机可能会失败，而使用被动探测检测到网络状态。

7.检查“MaxActiveProbes”值

NCSI 发送的活动探测数不会超过设置的值 `HKLM\SOFTWARE\Policies\Microsoft\Windows\NetworkConnectivityStatusIndicator\MaxActiveProbes`。

默认情况下，此键不存在（或设置为 *0*）。 这表示探测数量不受限制，这很少会引发问题。

8.检查 DNS 解析

对于 DNS 探测，请执行 nslookup 或 Resolve-DnsName ， `dns.msftncsi.com` 以确保可从计算机解析终结点。

9. 被动探测相关设置

检查注册表 `HKLM\SYSTEM\CurrentControlSet\Services\NlaSvc\Parameters\Internet\MinimumInternetHopCount` 值 。

值 *3* 通常适用于大多数企业基础结构。 默认值为 *8*。

检查组策略对象（GPO）配置：**计算机配置\管理模板\\网络连接状态指示器\指定被动轮询。**

> ⓘ **注意**
>
> 如果未配置组策略，则默认允许它。

常见问题和解决方案

由于探测服务器不正确，DNS 探测失败

确保注册表中为活动探测指定的 DNS 完全限定域名与目标探测服务器匹配。

除非需要，否则不建议修改默认探测数据配置。 虽然客户可以重写它以定向到其专用托管的探测服务器，但这种情况很少见。

DNS 探测由于超时而失败

了解为什么没有 DNS 响应。 可能的原因包括未检测到的代理和不正确的 DNS 服务器 IP 地址。

由于“HTTP 403 禁止访问”错误，HTTP 探测失败

检查环境中是否有任何防火墙或网关块。

DNS 解析无法发送 HTTP 探测

DNS 服务器应能够响应 `msftconnecttest.com` 查询或将其转发到可以执行此操作的服务器。

HTTP 直接探测失败，但代理存在

确保客户端收到对 Web 代理自动发现（WPAD）查询的有效 DNS 响应。 如果 DNS 服务器无法解析查询，则应将查询转发到可以执行此操作的 DNS 服务器。

其他选项包括手动配置代理数据（通过 GPO），或允许通过上游防火墙进行活动探测。

数据收集

在联系Microsoft支持人员之前，可以收集有关问题的信息。

先决条件

1. TroubleShootingScript （TSS） 必须由在本地系统上具有管理员权限的帐户运行，并且必须接受最终用户许可协议（EULA）（一旦接受 EULA，TSS 不会再次提示）。
2. 建议使用本地计算机 `RemoteSigned` PowerShell 执行策略。

> **注意**
>
> 如果当前 PowerShell 执行策略不允许运行 TSS，请执行以下操作：
>
> - 通过运行 cmdlet `Set-ExecutionPolicy -scope Process -ExecutionPolicy RemoteSigned` 设置 `RemoteSigned` 进程级别的执行策略。
> - 若要验证更改是否生效，请运行 cmdlet `Get-ExecutionPolicy -List`。
> - 由于进程级别权限仅适用于当前的 PowerShell 会话，因此一旦关闭了运行 TSS 的给定 PowerShell 窗口，进程级别的分配权限也将返回到以前配置的状态。

在联系Microsoft支持人员之前收集关键信息

1. 下载 TSS 并将其解压缩到 *C：\tss* 文件夹中。

2. *从提升的 PowerShell 命令提示符打开 C：\tss* 文件夹。

3. 使用以下 cmdlet 启动受影响计算机上的跟踪：

```powershell
.\TSS.ps1 -Start -Scenario NET_NCSI
```

4. 接受 EULA 和问题步骤记录器（PSR）确认。

5. 在输入 *Y* 之前重现问题。

6. 在重现问题后输入 *Y* 以完成日志收集。

跟踪将存储在 C：\MS_DATA *文件夹中的* zip 文件中。

常见问题解答

问题 1. 何时发送主动探测?

活动探测由某些事件触发。 NCSI 监视器或已注册以接收指示可能需要刷新网络状态的事件通知。

问题 2. NCSI 如何知道是要使用 HTTP 还是 DNS 探测?

- 如果不存在代理，则使用 DNS 进行 NCSI 探测。
- 如果检测到代理，NCSI 将使用 HTTP 探测。
- 当代理的存在尚未得到确认时，还有一个“强制”的 Web 探测。 有时，NCSI 可能会发现代理。 但与此同时，如果 DNS 探测不起作用，它可能会怀疑代理没有明确的证据。 在这两种情况下，NCSI 都使用 HTTP 探测。
- Wi-Fi 和 IPv6 接口始终使用 HTTP 探测。

问题 3. 被动探测何时运行以及多久运行一次?

仅当满足这些条件时，被动探测才会运行：

- 组策略允许以下设置。

  计算机配置\管理模板\\网络连接状态指示器\指定被动轮询

  > ⓘ **注意**
  >
  > 如果未配置组策略，则默认允许它。

- 至少有一个客户端应用程序或服务已注册 NCSI 通知。

- 用户已登录或在过去 30 秒内登录。

- 系统未在网络安静模式下运行。

- 接口上存在 IPv4/IPv6 单播地址，在过去 30 秒内处理/或数据包。

问 4. 为什么同时需要主动和被动探测?

主动和被动探测相互补充。 它们都确定相同的最终结果（连接类型），但以不同的方式。 由于网络状况间歇性，这两者都需要它们。 有时，这些条件会阻止活动探测正常运行。

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